

LESSON 2

Probability

1. Basic concepts

2. Events

2.1. Types of events

2.2. Operations with events

3. Probability calculation

3.1. Laplace rule

3.3. Probability of the union of the events

4. Conditional probability

4.1. Intersection theorem

5. Event independence

6. Theorem of total probability

7. Bayes' Theorem

Probability inference

The probability calculation is the basic mathematical model on which is built the Statistical Inference.

Random experiment

An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a random experiment. For example: rolling a die

Deterministic experiment

Whenever it is performed under the **same conditions** produces the **same result**.
For exmple: Measure the time of a falling object with a certain weight from a specific height.

Random experiment

An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a random experiment. For example: rolling a die

Population

In the terminology statistics is called population to the set of all them "individuals" or entities that constitute the object of a specific study and on which is wanted to get certain conclusions.

Random variable

A random variable can be defined as the result of an experiment or a random experience, and can be coded numerically.

Sample Space (E)

Set of all possible outcomes in random experience that we are doing; rather, is the joint formed by all the possible values or results of the random variable that we are considering. It is represented with the letter E.

Outcomes from rolling a die: $E=\{1,2,3,4,5,6\}$

Events definition

Any result that, objectively, we can say if it has happened or not, at the end of the random experience. An event is a subset of the sample space of a random experiment.

Outcomes from rolling a die.

$$E=\{1,2,3,4,5,6\}$$

$$A=\{2,4,6\} = \text{To get an even number}$$

Properties:

- Events are denoted in capital letters (A, B, C, etc.).
- The elements of any event A, viewed as a whole, are the different results that comprises or contains.
- If performing a random experience, an element of the event A is observed, we say that event A has occurred. Otherwise, we say that it has not happened.

Types of events – Example rolling a die

Impossible event ($\emptyset$)

An event which cannot occur at any performance of the experiment

Certain event (A)

An event which is sure to occur at every performance of an experiment.

Mutually exclusive events (A, B)

These events cannot occur at the same time. They do not have any element in common.

Complementary event ($\bar{A}$)

An event which consists in the negation of another event. Not happening of the event A is called the complementary event of the event A

Types of events – Example rolling a die

Impossible event ($\emptyset$)

$$\emptyset = \{7\}$$

Certain event (A)

$$A = \{\text{number less or equal to 6}\}$$

Mutually exclusive events (A, B)

$$A = \{1, 2\} \text{ and } B = \{3, 4\}$$

Complementary event ($\bar{A}$)

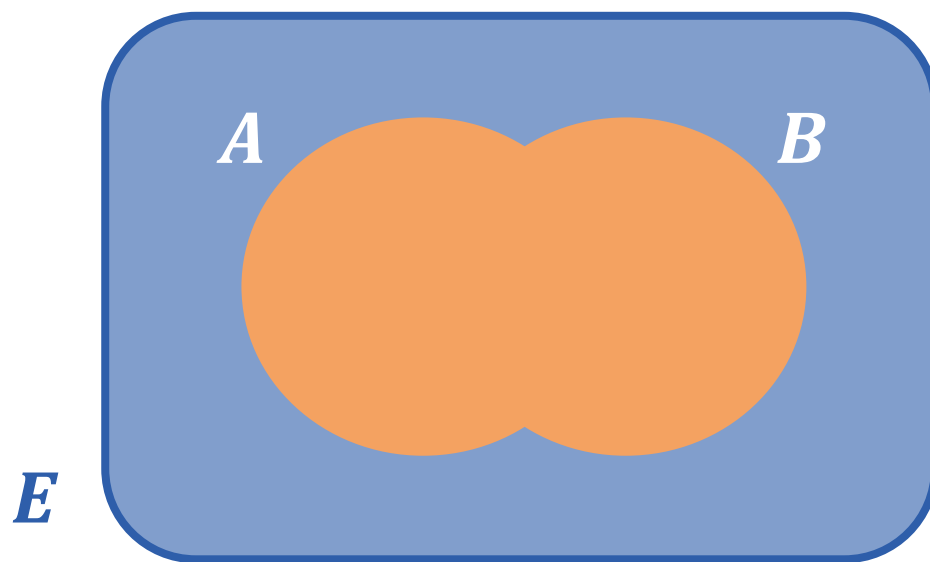
$$A = \{1, 3, 5\} \text{ and } \bar{A} = \{2, 4, 6\}$$

Operations with events

Union

The union of two events is the event that consists of all outcomes that are contained in either of the two events

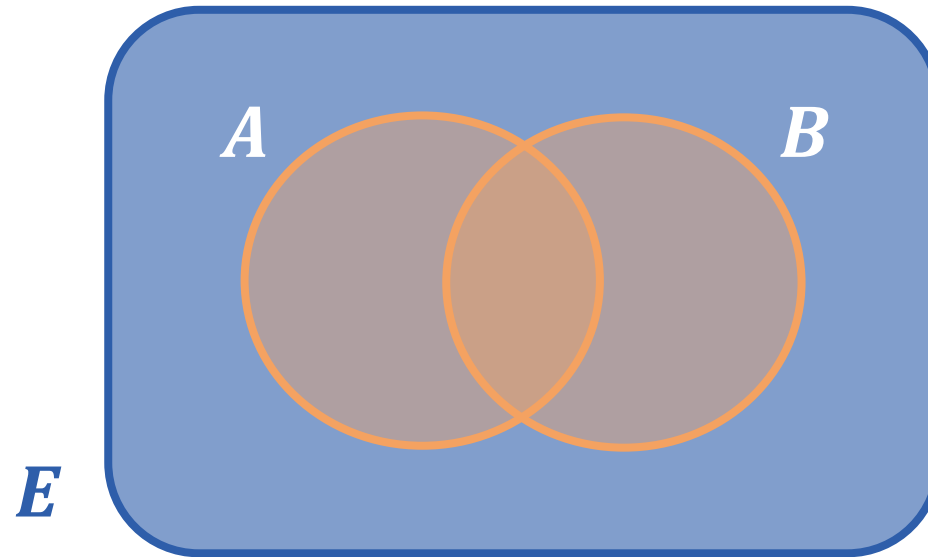
$$A \cup B$$



Intersection

The intersection of two events is the event that consists of all outcomes that are contained in both of the two events.

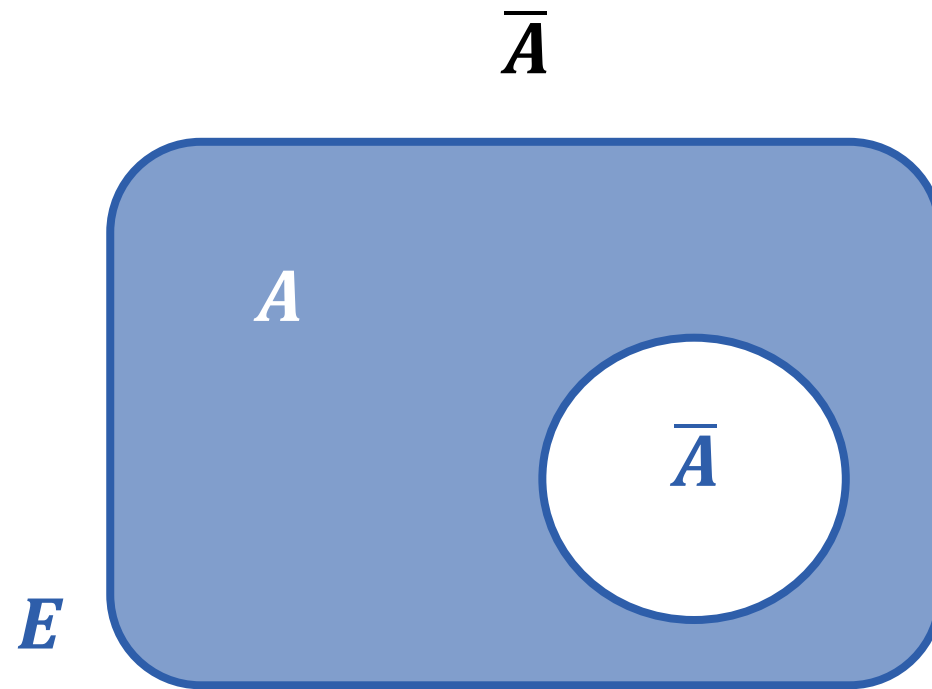
$$A \cap B$$



Operations with events

Complement event

The complement of an event in a sample space is the set of outcomes in the sample space that are not in the event.



$$\bar{A} \cup A = E$$

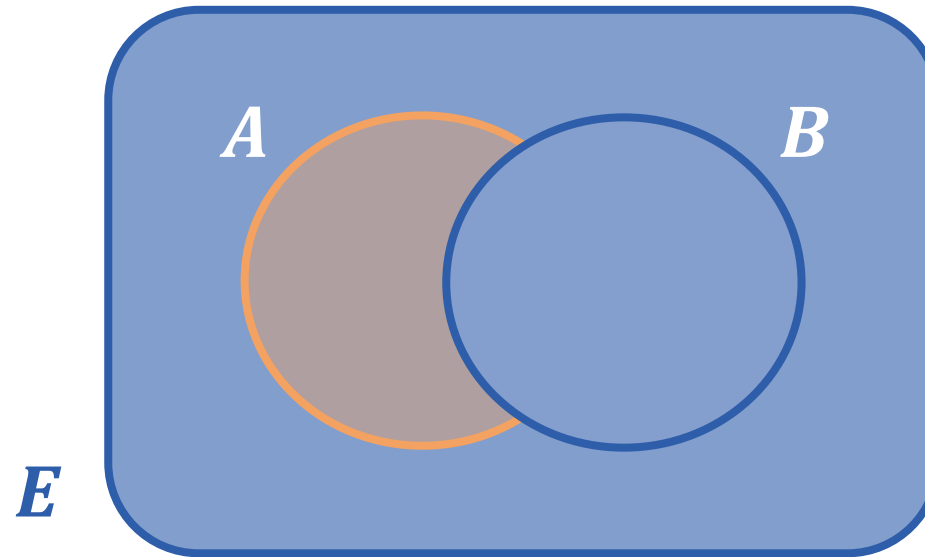
$$\bar{A} \cap A = \emptyset$$

Operations with events

Difference event

The difference of two probabilistic events A and B is the event consisting of all outcomes that occur in A but not in B

$$A - B$$



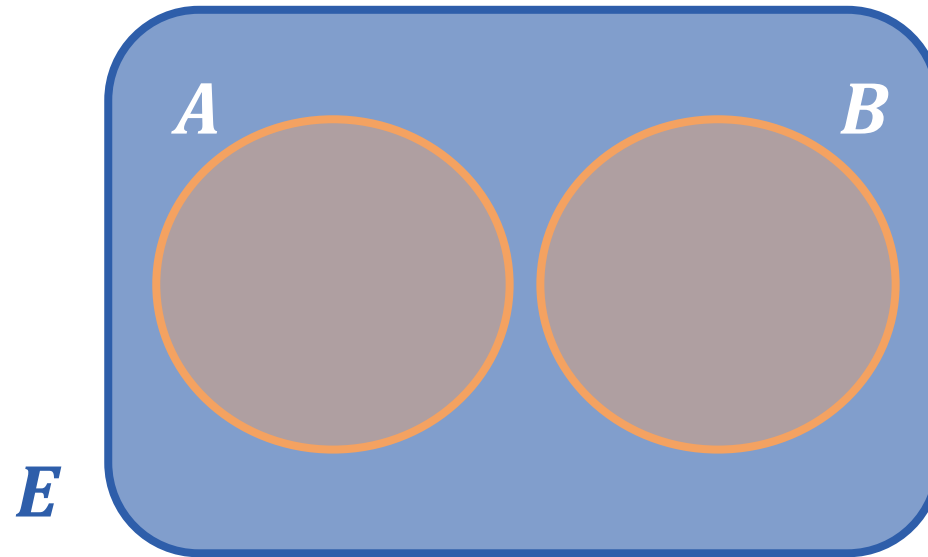
$$A - B = A \cap \bar{B}$$

Operations with events

Mutually exclusive events

Two events are said to be mutually exclusive if they can not occur at the same time

$$A \cap B = \emptyset$$



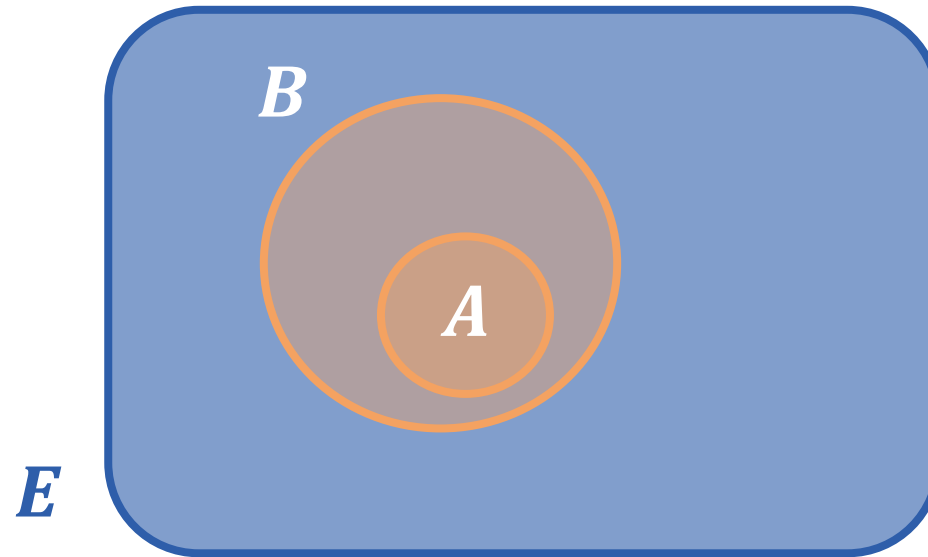
Operations with events

Contained events

An event A is contained in an event B if every outcome that belongs to A also belongs to B .

I

$$A \subseteq B \text{ or } A \rightarrow B$$

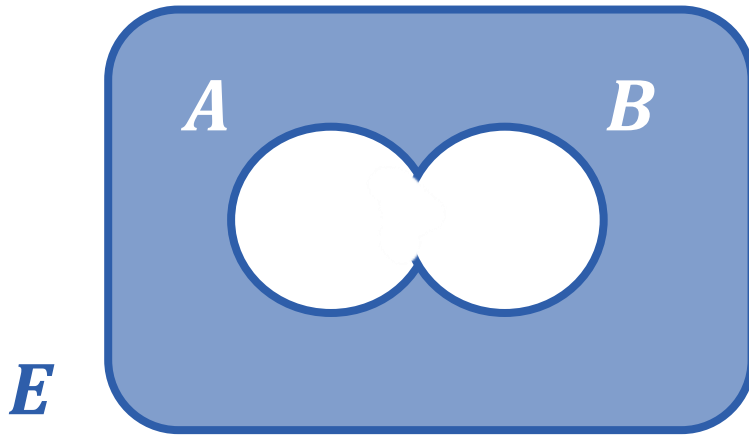


Operations with events

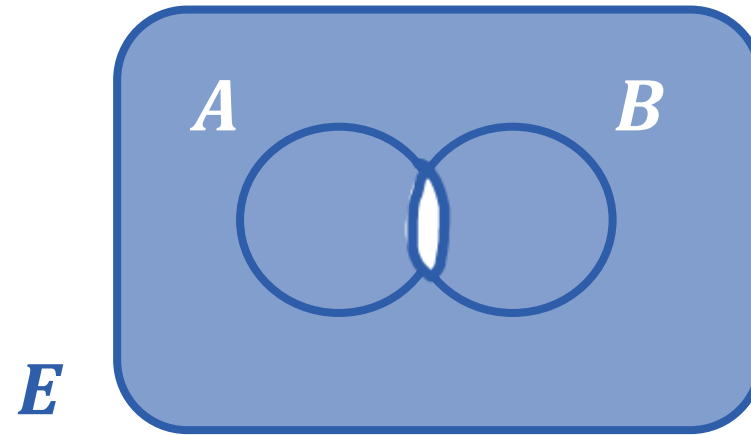
Morgan Law's

The complement of the union of two events equals the intersection of their complements, and the complement of their intersection equals the union of their complements.

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$



$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$



Probability calculation – Laplace rule

Definition

A numerical measure of how likely an event is to occur, taking values between 0 and 1, where 0 means impossible and 1 means certain.

Laplace rule

If we have a random experiment with a finite number of possible outcomes and all the outcomes are equally likely (for example, rolling a dice, a coin, remove a card from a deck, etc.) it can be verified that:

$$P(A) = \frac{\text{Favourable cases}}{\text{Possible cases}} = \frac{n^{\circ} \text{ of outcomes in } A}{n^{\circ} \text{ of outcomes in } E}$$

Probability calculation – Properties

- $P(A) \geq 0$ and $P(A) \leq 1, \forall A \in E$
- $P(E) = 1$
- $P(\emptyset) = 0$
- $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive
- $P(\bar{A}) = 1 - P(A), \forall A \in E$
- If $A \subseteq B \rightarrow P(A) \leq P(B)$

Probability of the union of the events

It has been seen in one of the fundamental properties that, if A and B are mutually exclusive events, then:

$$P(A \cup B) = P(A) + P(B)$$

If A and B are any events, then it can be demonstrated the following property, very useful and important:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If we have 3 events A, B and C, the probability of their union are:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(C \cap B) + P(A \cap B \cap C)$$

Conditional probability

We call probability of A conditioned by the event B to the probability of A, knowing that the event B has occurred (ie, in the presence of event B), and is expressed as

$$P(A|B)$$

We are calculating the probability of event A. The event B is supposed to "has already happened."

Conditional probability – Intersection theorem

Obtaining the probability of A knowing B is the same as if we consider only the part of A that is inside B ($A \cap B$), because the part of A outside B cannot occurred and therefore it is the same as calculating the proportion of $A \cap B$ with respect to B.

If we know or we can calculate the probability of B and the probability of $A \cap B$, therefore we can calculate $P(A/B)$ with the following expression:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Event independence

If the probability of an event does not change in the presence or absence of another event, we can say that they are independent events:

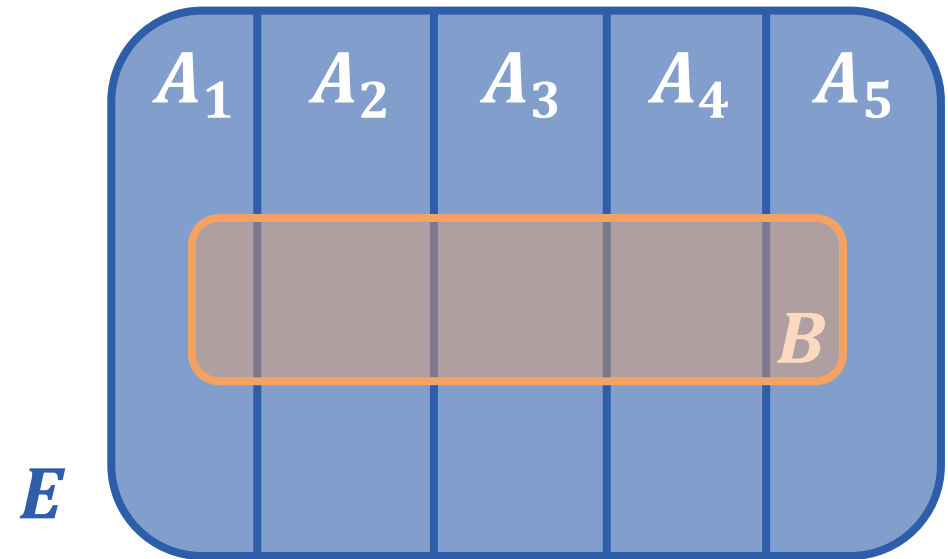
$$P(A|B) = P(A)$$

In the case of independent events (and only in this case) it is satisfied:

$$P(A \cap B) = P(A) \cdot P(B)$$

Theorem of total probability

Sometimes we do not know or we cannot directly calculate the probability of an Event **B**, however, we might be able to calculate its probability conditioned to different A_i events that partition the Sampling Space.



Theorem of total probability

Given a sampling space E , a partition is a set of events A_i that verifies the following:

- Are mutually exclusive events:

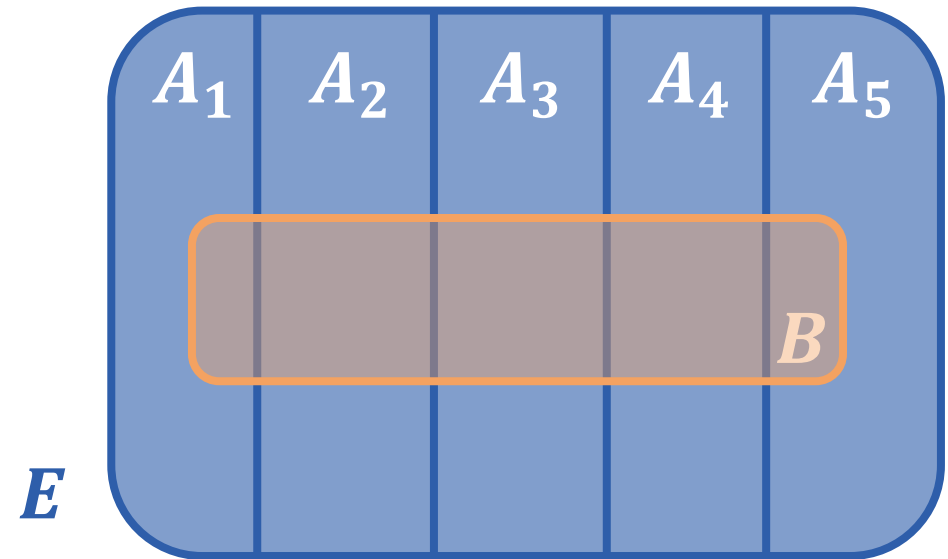
$$A_i \cap A_j = \emptyset, \quad \forall i \neq j$$

- The union of the events is equal to E

$$A_1 \cup A_2 \cup \dots \cup A_n = E$$

Any event B can be expressed as:

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)$$



Theorem of total probability

We can "decompose" the probability of event **B** in the sum of the probabilities of each "piece" in which the event is divided:

$$P(B) = P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + \cdots + P(B|A_n) \cdot P(A_n)$$

Bayes' Theorem

Bayes' Theorem is used when it is known $P(B/A)$ but not $P(A/B)$, solving the problem of the inverse probability and allowing to calculate the probability that knowing that effect B has occurred, it was because of one of the causes A_i :

$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{P(B)}$$

Applying the total probability theorem:

$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + \dots + P(B|A_n) \cdot P(A_n)}$$